Supplementary Table S1. Qualitative assessment of study reporting

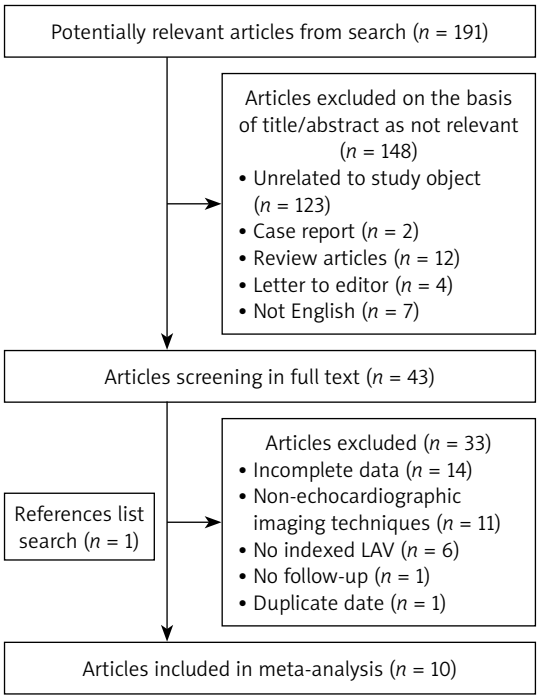
Domain	Questions	Judgments
Risk of bias		
1) Patient selection	Was a consecutive or random sample of patients enrolled?	Yes, No, Unclear
	Was a case-control design avoided?	Yes, No, Unclear
	Did the study avoid inappropriate exclusions?	Yes, No, Unclear
	Could the selection of patients have introduced bias?	Low, High, Unclear
Applicability	Is there concern that the included patients do not match	Low, High, Unclear
1) Patient selection	the review questions?	
Risk of bias	Were the index test results interpreted without knowledge	
2) Index test	of the results of the reference standard?	Yes, No, Unclear
	If a threshold was used, was it pre-specified?	Yes, No, Unclear
	Could the conduct or interpretation of the index test have	
	introduced bias?	Low, High, Unclear
Applicability	Is there concern that the index test, its conduct, or	
2) Index test	interpretation differ from the review question?	Low, High, Unclear
Risk of bias	Is the reference standard likely to correctly classify the	
3) Reference standard	target condition?	Yes, No, Unclear
	Were the reference standard results interpreted	
	without knowledge of the results of the index test?	Yes, No, Unclear
	Could the reference standard, its conduct, or its interpretation	
	have introduced bias?	Low, High, Unclear
Applicability	Is there concern that the target condition as defined by the	
3) Reference standard	reference standard does not match the review question?	Low, High, Unclear
Risk of bias	Was there an appropriate interval between index test(s)	
4) Flow and timing	and reference standard?	Yes, No, Unclear
	Did all patients receive a reference standard?	Yes, No, Unclear
	Did patients receive the same reference standard?	Yes, No, Unclear
	Were all patients included in the analysis?	Yes, No, Unclear
	Could the patient flow have introduced bias?	Low, High, Unclear

Patients – Patients undergo CRT; Criteria for CRT. Index test – Relationship between CRT responders and LAVI. Comparator test (if applicable) – Mean change of LAVI on related mean change of LVESV and/or LVEF. Target condition – Predictive value of LAVI; Reference standard: Guidelines of heart failure and indications of cardiac resynchronisation therapy in heart failure.

Supplementary Table SII. Summary of QUADAS-2 Assessment of Selected Studies

Author, year [reference]	Risk of bias				Applicability concerns		
	Patients selection	Index test	Reference standard	Flow and timing	Patients selection	Index test	Reference standard
Marsan, 2008 [22]	Low	Low	Low	Low	Low	Low	Low
Donal, 2009 [23]	Low	Low	Low	Unclear	Low	Low	Low
Shanks, 2011 [24]	Low	High	Low	Low	Low	High	Unclear
Hsu, 2012 [25]	High	Low	Low	Low	High	Unclear	Low
Imamura, 2014 [26]	Low	High	Unclear	Unclear	Low	High	Unclear
Feneon, 2015 [27]	Low	Low	Low	Low	Low	Low	Low
van 't, 2015 [28]	Unclear	Low	Low	Low	Unclear	Low	Low
Kloosterman, 2016 [29]	Low	Low	Low	Low	Low	Low	Low
Badran, 2017 [30]	Low	Low	Low	Low	Low	Low	Low
Hansen, 2017 [31]	Low	Low	Low	Low	Low	Low	Low

QUADAS-2 – Quality Assessment of Diagnostic Accuracy Studies-2.



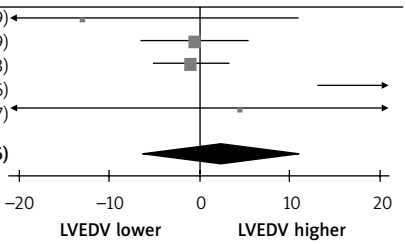
Supplementary Figure S1. Flow chart of study selection

A

Study or subgroup	Response Mean	SD	Total	Non-response Mean	SD	Total	Weight (%)	Mean difference, IV, random, 95% CI
Marsan 2008	195	51	34	208	35	17	10.0	–13 (–36.89, 10.89)
Shanks 2011	107.8	36.9	327	108.4	36.3	254	35.7	–0.60 (–6.59, 5.39)
Hsu 2012	123	26	191	124	25	562	39.1	–1.00 (–5.23, 3.23)
Kloosterman 2016	254	107	201	220	97	164	12.1	34.00 (13.04, 54.96)
Badran 2017	255.5	77	24	251.13	67	13	3.0	4.37 (–43.33, 52.07)
Total (95% CI)			**777**			**1010**	**100.0**	**2.35 (–6.25, 10.96)**

Heterogeneity: $\tau^2 = 44.60$; $\chi^2 = 11.52$, d$f = 4$ ($p = 0.02$); $I^2 = 65\%$

Test for overall effect: $Z = 0.54$ ($p = 0.59$)

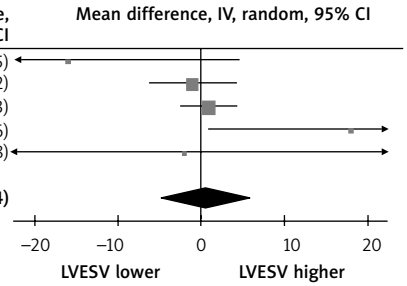


B

Study or subgroup	Response Mean	SD	Total	Non-response Mean	SD	Total	Weight (%)	Mean difference, IV, random, 95% CI
Marsan 2008	140	44	34	156	30	17	6.0	–16.00 (–36.55, 4.55)
Shanks 2011	80.6	33	327	81.6	32	254	37.0	–1.00 (–6.32, 4.32)
Hsu 2012	88.5	21	191	87.6	20.6	562	47.2	0.90 (–2.53, 4.33)
Kloosterman 2016	187	87	201	169	80	164	8.2	18.00 (0.84, 35.16)
Badran 2017	192	67	24	194	58	13	1.6	–2.00 (–43.38, 39.38)
Total (95% CI)			**777**			**1010**	**100.0**	**0.54 (–4.75, 5.84)**

Heterogeneity: $\tau^2 = 12.40$; $\chi^2 = 6.84$, d$f = 4$ ($p = 0.14$); $I^2 = 42\%$

Test for overall effect: $Z = 0.20$ ($p = 0.84$)

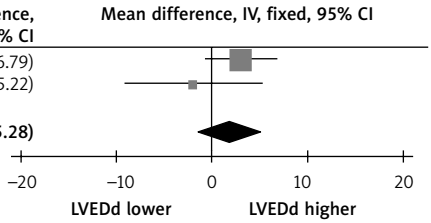


C

Study or subgroup	Response Mean	SD	Total	Non-response Mean	SD	Total	Weight (%)	Mean difference, IV, fixed, 95% CI
Fenon *et al.*, 2015	69	8	54	66	8	25	78.4	3.00 (–0.79, 6.79)
Imamura 2014	69	11	11	71	12	56	21.6	–2.00 (–9.22, 5.22)
Total (95% CI)			**65**			**81**	**100.0**	**1.92 (–1.44, 5.28)**

Heterogeneity: $\chi^2 = 1.44$; d$f = 1$ ($p = 0.23$); $I^2 = 31\%$

Test for overall effect: $Z = 1.12$ ($p = 0.26$)

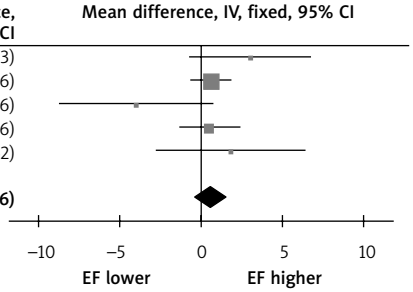


D

Study or subgroup	Response Mean	SD	Total	Non-response Mean	SD	Total	Weight (%)	Mean difference, IV, fixed, 95% CI
Marsan 2008	29	5	34	26	7	17	6.7	3.00 (–0.73, 6.73)
Shanks 2011	26.2	7.6	327	25.6	7.8	254	58.0	0.60 (–0.66, 1.86)
Imamura 2014	21	7	11	25	9	56	4.1	–4.00 (–8.76, 0.76)
Kloosterman 2016	24	9	201	23.5	9	164	26.9	0.50 (–1.36, 2.36)
Badran 2017	25.44	7.4	24	23.63	6.5	13	4.4	1.81 (–2.80, 6.42)
Total (95% CI)			**597**			**504**	**100.0**	**0.60 (–0.36, 1.56)**

Heterogeneity: $\chi^2 = 5.45$; $\chi^2 = 4$, d$f = 4$ ($p = 0.24$); $I^2 = 27\%$

Test for overall effect: $Z = 1.22$ ($p = 0.22$)

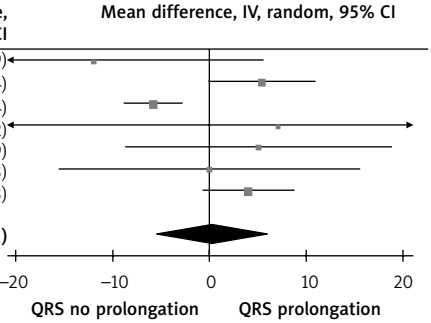


Supplementary Figure S2. Comparison of baseline LVEDV (**A**), LVESV (**B**), LVEDd (**C**), and EF (**D**) in the group of patients with CRT response vs. CRT non-response

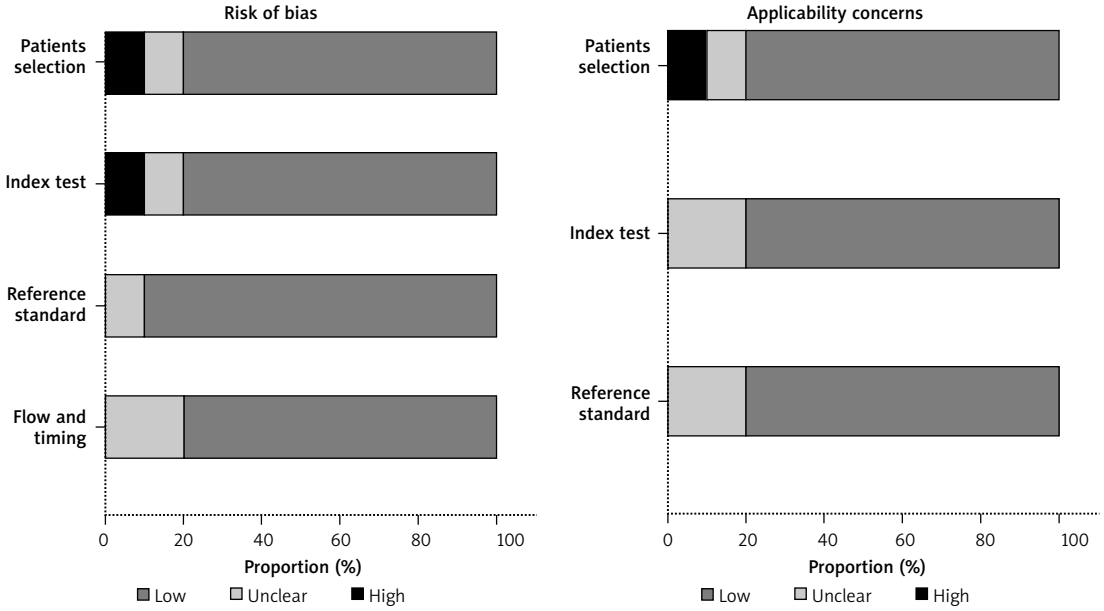
Study or subgroup	Response Mean	SD	Total	Non-response Mean	SD	Total	Weight (%)	Mean difference, IV, random, 95% CI
Marsan 2008	142	28	34	154	31	17	7.7	–12.00 (–29.49, 5.49)
Shanks 2011	156	32.5	237	150.6	29.9	254	21.6	5.40 (–0.14, 10.94)
Hsu 2012	153.9	18.1	191	159.7	20.1	562	25.0	–5.80 (–8.86, –2.74)
Imamura 2014	148	49	11	141	33	56	3.2	7.00 (–23.22, 37.22)
Feneon *et al.* 2015	163	27	54	158	30	25	10.6	5.00 (–8.79, 18.79)
Van't 2015	160	26	63	160	25	12	9.1	0.00 (–15.53, 15.53)
Kloosterman 2016	162	24	201	158	22	164	22.8	4.00 (–0.73, 8.73)
Total (95% CI)			**791**			**1090**	**100.0**	**0.45 (–5.31, 6.21)**

Heterogeneity: $\tau^2 = 32.03$, $\chi^2 = 21.33$; $\chi^2 = 4$, d$f = 6$ ($p = 0.002$); $I^2 = 72\%$

Test for overall effect: $Z = 0.15$ ($p = 0.88$)



Supplementary Figure S3. Comparison of baseline QRS duration in group of patients with CRT response vs. CRT non-response



Supplementary Figure S4. Summary of QUADAS-2 Assessment of Selected Studies